

Question Paper

Exam Date & Time: 01-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (INFORMATION TECHNOLOGY) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ICT 3121/ICT_3121 - INFORMATION SECURITY
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Given the hex code of the plaintext {54 77 6F 20 4F 6E 65 20 4E 69 6E 65 20 54 77 6F} and the initial key {54 68 61 74 73 20 6D 79 20 4B 75 6E 67 20 46 75} answer the following by applying the functions of Advanced Encryption Standard. Refer to the tables 1A (a) and 1A (b). (5)
- Show the original State displayed as 4X4 matrix and the value after initial add round key.
 - Show the value of the State after SubBytes.
 - Show the value of the State after ShiftRows.
 - Using Key Expansion method compute the complete first round key W_4 - W_7 for the initial key stream given above.

Table 1A(a) : RCON Constants

Round	Constant (RCon)	Round	Constant (RCon)
1	(01 00 00 00) ₁₆	6	(20 00 00 00) ₁₆
2	(02 00 00 00) ₁₆	7	(40 00 00 00) ₁₆
3	(04 00 00 00) ₁₆	8	(80 00 00 00) ₁₆
4	(08 00 00 00) ₁₆	9	(1B 00 00 00) ₁₆
5	(10 00 00 00) ₁₆	10	(36 00 00 00) ₁₆

Table 1A(b) : Sub Bytes

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

- 1B) A company wants to secure its communication by using both steganography and cryptography. (3)

Apply your understanding of these techniques to propose a solution where both methods are used together. Explain how each technique enhances security and describe a potential scenario where this combined approach could be effective.

1C) Employ AES-192 key Expansion to obtain W_6 and W_7 for the key stream (2)
 $\{0x8E, 0x73, 0xB0, 0xF7, 0xDA, 0x0E, 0x64, 0x52, 0xC8, 0x10, 0x53, 0x2B, 0x80, 0x90, 0x79, 0xE5, 0x62, 0xF8, 0xEA, 0xD2, 0x52, 0x2C, 0x6B, 0x7B\}$.

2A) Raj uses ElGamal for creating a secure system which projects Confidentiality, Integrity, Nonrepudiation and Authentication. Given the prime $p=13$, $e_1 = 2$, $r=5$ and $d=3$. (5)

I) Encrypt the message "11".

II) Decrypt the cipher text to get back the plain text.

III) Calculate the two signatures S_1 and S_2 .

IV) Show the verification process for the above calculated signatures.

Show with suitable justifications how Confidentiality, Integrity, Nonrepudiation and Authentication is maintained in the above communication scenario.

2B) Alice and Bob were two cryptographers tasked with establishing a secure communication channel across an unsafe network using the Diffie-Hellman key exchange protocol. They employ the prime number $p: 97$ and the Generator number $\alpha: 2$. The private keys in question are Bob's private key $b: 27$ and Alice's private key $a: 13$. Calculate their common secret key. (3)
 Also, justify why Diffie-Hellman key exchange is more secure than traditional methods of key exchange

2C) Evaluate the trade-offs between HMAC-SHA256 and HMAC-SHA1 in terms of security, efficiency, and output size. In which scenarios would HMAC-SHA256 be a more suitable choice? (2)

3A) Distinguish between preimage and second preimage resistance with suitable examples. (5)
 Compute the following in reference to SHA512:
 I) What is the leftmost digit of the result, when majority function is applied to buffers A, B and C with initial contents $0x7$, $0xA$ and $0xE$ respectively.
 II) What is the leftmost digit of the result, when conditional function is applied to buffers E, F and G with initial contents $0x9$, $0xA$ and $0xF$ respectively.

3B) Alice chooses $q = 101$ and $p = 8081$. Alice selects $e_0 = 3$ and calculates $e_1 = 6968$. Alice also selects $d = 61$ as private key and calculates $e_2 = 2038$. Given that $h(M) = 6000$ and Alice chooses $r = 61$, demonstrate how Bob accepts the message using the Digital Signature Standard (DSS). (3)

3C) Given an elliptic curve and two points $P = (2, 5)$ and $Q = (4, 12)$ on the curve, calculate the point $P + Q$ using the group law. The Elliptic Curve: $y^2 = x^3 + 2x + 2 \pmod{17}$. (2)

4A) Discuss the various approaches to challenge response authentication using symmetric key encryption. (5)

4B) Let message = "graduate", Key = "word", find ciphertext using Playfair cipher (3)

4C) A streaming service wants to protect its video content from unauthorized access and piracy. They decide to implement encryption for both the video files stored on their servers and the streaming process. (2)
 I) What encryption techniques could the streaming service use to secure their video content?
 II) How can they manage encryption keys to ensure that only authorized users can access the videos?

What are some potential challenges they might face in implementing multimedia encryption?

5A) Evaluate the different protocols of SSL. Explain Handshake protocol in detail. Describe the SSL Architecture in detail with a neat diagram. (5)

5B) Consider a Simple LAN consisting of a number of clients, a server, a mail server and Internet access. A firewall is designed to divide this LAN into three isolated segments, namely internet segment, client segment and server segment. We are writing the rules to allow only the required service to be accessed from one segment to other. (3)
 Our firewall is now configured to allow only http requests to the web server and no access at all from outside to our internal LAN.
 i) Does this mean that web server is secure

ii) Is the rest of the internal LAN secure
Justify your answer and suggest ideas how to ensure security to the given scenario.

5C) Differentiate between Searchable encryption and Homomorphic encryption. (2)

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